

Frog Rectus Abdominis CAL Simulator

User Manual · Version 1.0

Computer-Assisted Learning Software
MBBS Phase II · Experimental Pharmacology

Department of Pharmacology

NMC CBME Compliant · 3Rs Replacement · Offline Desktop
Application

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1. Introduction

The **Frog Rectus Abdominis CAL Simulator** is a desktop application designed for MBBS Phase II pharmacology practical teaching. It replaces the traditional animal experiment with a computer-based simulation that teaches the same core pharmacological concepts.

What students will learn:

- Drug action on skeletal (nicotinic) receptors
- Dose-response relationships and the Hill equation
- Graded muscle contraction in an organ bath
- Competitive antagonism and rightward DRC shift
- Potency (EC_{50}) and efficacy (E_{max}) comparison
- Unknown drug identification by pharmacological profile

System Requirements:

Item	Requirement
OS	Windows 10/11, macOS 12+, or Linux
Python	3.9 or higher (for source code)
RAM	4 GB minimum
Disk	~100 MB (standalone EXE)
Display	1280 x 720 minimum
Network	Not required (fully offline)

2. Installation

Option A: Run from Source Code

1. Install Python 3.9+ from python.org
2. Open Command Prompt / Terminal
3. Navigate to the frog_cal folder: `cd frog_cal`
4. Install dependencies: `pip install -r requirements.txt`
5. Run the application: `python main.py`

Option B: Standalone EXE (Windows)

1. Double-click `FrogCAL_Simulator.exe`
2. No installation or Python required
3. Windows may show a security warning on first run — click 'More Info' then 'Run anyway'

3. Home Screen Overview

The home screen provides navigation buttons to all modules. Click any button to enter that module. Each module has a 'Back' button to return to the home screen.

Module	Description
Theory	Illustrated notes on frog rectus abdominis pharmacology
Virtual Experiment Lab	Interactive organ bath with live contraction tracing
Dose-Response Curve	Automatic DRC plotting on semi-log scale
Antagonism Study	Competitive antagonism with rightward shift demo
Unknown Drug Demo	Identify unknown drugs by comparing DRC profiles
Settings	Sound, fullscreen, animation speed configuration

4. Theory Module

The Theory module contains seven illustrated sections covering the scientific background of the frog rectus abdominis preparation. Select any topic from the sidebar to read the content.

Topics covered:

- Frog Rectus Abdominis – Introduction
- Historical Importance (Burn's bioassay)
- Acetylcholine Effect (nicotinic mechanism)
- Receptor Interaction (Occupation Theory)
- Dose-Response Relationship (Hill equation)
- Competitive Antagonism (Schild equation)
- 3Rs – Ethical Replacement & NMC CBME

5. Virtual Experiment Lab

This is the core module. It simulates an organ bath experiment with a frog rectus abdominis muscle strip.

Layout

The screen is divided into three panels: **Controls** (left), **Organ Bath** (centre), and **Kymograph** (right).

Step-by-Step Procedure

Step 1: Select an agonist (e.g. Acetylcholine) from the dropdown

Step 2: Set the dose in micromolar (e.g. 1.0 μM)

Step 3: Click 'Add Drug to Bath' — watch the muscle contract and the graph trace rise

Step 4: Click 'Wash Tissue' to remove the drug — muscle relaxes back to baseline

Step 5: Increase the dose and repeat to see graded responses

Step 6: (Optional) Add an antagonist before the agonist to see reduced response

Note: The experiment log (bottom-left) records every action. The organ bath shows real-time muscle contraction, solution colour, aeration bubbles, and drug presence.

Available Drugs

Drug	Type	EC50	E _{max}
Acetylcholine	Agonist	3 μM	100%
Carbachol	Agonist	1 μM	100%
Nicotine	Agonist	5 μM	85%
Succinylcholine	Agonist	8 μM	95%
d-Tubocurarine	Antagonist	KB=200 nM	N/A
Atropine	Antagonist	KB=1 nM	N/A

6. Dose-Response Curve Module

This module automatically generates dose-response curves on a semi-log scale. It calculates responses across a range of concentrations using the Hill equation and plots them.

How to use:

1. Select a drug from the dropdown
2. Click 'Plot DRC' to generate the curve
3. Click 'Add Another Drug' to overlay multiple curves for comparison
4. Toggle EC_{50} lines and E_{max} lines using checkboxes
5. Compare potency by examining the horizontal position of curves

Teaching point: A curve shifted to the LEFT indicates HIGHER potency (lower EC_{50}). A curve with lower maximum indicates lower efficacy. Carbachol is more potent than ACh (leftward curve) because it is not degraded by acetylcholinesterase.

7. Antagonism Study

This module demonstrates competitive antagonism visually. It shows how a competitive antagonist shifts the agonist DRC to the right without reducing the maximum response.

How to use:

1. Select agonist and antagonist
2. Click 'Plot Control DRC' to see the baseline curve
3. Set an antagonist concentration and click 'Add Shifted DRC'
4. Observe the rightward parallel shift
5. Use 'Auto: Show 3 Shifts' for a complete demonstration

Key observation: In competitive antagonism, the $E_{\max}$ remains unchanged (the curves reach the same maximum). The Dose Ratio (DR) follows the Schild equation: $DR = 1 + [B]/K_B$.
d-Tubocurarine is the classical competitive antagonist at nicotinic receptors.

8. Unknown Drug Demo

This module tests the student's ability to identify an unknown drug based on its dose-response profile.

1. Click 'Generate Unknown Drug' — a random agonist is selected
2. The unknown DRC appears in black on the plot
3. Add reference curves of known drugs using the dropdown
4. Compare EC_{50} and E_{max} to identify the unknown
5. Select your answer and click 'Submit Answer' for instant feedback

9. Settings

Setting	Options	Default
Fullscreen	On / Off	Off
Sound Effects	On / Off	On
Animation Speed	0.5x to 2.0x	1.0x

10. Pharmacology Engine Details

Hill Equation

The drug response is calculated using the Hill equation:

$$E = (E_{\max} \times D^n) / (EC50^n + D^n)$$

Where E = response, D = drug concentration, EC50 = concentration giving 50% response, n = Hill coefficient.

Competitive Antagonism (Schild Equation)

In the presence of a competitive antagonist, the apparent EC50 shifts:

$$EC50' = EC50 \times (1 + [Antagonist] / KB)$$

Biological Variability

A +/-3% Gaussian random noise is added to each response to simulate biological variability. Tachyphylaxis (tissue fatigue) is modelled after 8+ consecutive stimulations.

Contraction Animation

The muscle contraction trace follows realistic kinetics: baseline, sigmoidal rise, peak plateau with tremor, and exponential washout decay.

11. Building Standalone EXE

To create a standalone Windows executable that requires no Python installation:

1. Install PyInstaller: `pip install pyinstaller`
2. Navigate to `frog_cal` folder
3. Run: `python build_exe.py`
4. The EXE will be in `dist/FrogCAL_Simulator.exe`
5. Distribute this single file to students

Tip: The EXE is fully self-contained. Students do not need Python, PySide6, or any other software installed on their computers.

12. Troubleshooting

Problem	Solution
App won't start	Ensure Python 3.9+ and PySide6 are installed. Run: pip install PySide6
Graph not updating	Click 'Wash Tissue' before adding next dose. Wait for current animation to finish.
EXE blocked by antivirus	Add exception for FrogCAL_Simulator.exe or run from source code instead.
Small text on HiDPI	Enable fullscreen mode in Settings. Or set Windows display scaling to 100%.
Animation too fast/slow	Adjust Animation Speed slider in Settings.